

C Programming Basics

1. What is C?

C is a general-purpose programming language used for system programming and application development.

2. Variables

Common data types: int (integer), float (decimal), char (character). Example: `int age = 20;`

3. Input and Output

`printf()` displays output. `scanf()` takes input.

Example: `scanf("%d", &age);`

4. Conditions

if and else are used to make decisions.

Example: `if (age >= 18) printf("Adult");`

5. Loops

for, while, and do-while loops repeat code.

Example: `for (int i = 0; i < 5; i++) printf("%d", i);`

6. Functions

Functions are reusable blocks of code.

Example: `int add(int a, int b) { return a + b; }`

7. Arrays

An array stores multiple values of the same type.

Example: `int numbers[3] = {10, 20, 30};`

8. Pointers

A pointer stores the memory address of another variable.

Example: `int *p = &x;`